

24CS31 — 3-HOUR HAIL MARY

Ordered by **worst-case marks per minute**, computed against all four past papers. Do them **in this exact order**. Every calculus-heavy question has been cut **except** the four that repeat verbatim.

THE ONE THING TO UNDERSTAND. The “running total” column is the mark you would score on the **worst** of the four past papers if you stopped studying at that row. You cross the pass mark at **2 h 09**. You do **not** get to stop before then — two hours of this leaves you at 42. **Steps 1-12 are the whole plan. Finish them.**

THE CLOCK

stop-anywhere score = worst of 4 papers

#	Study this	Unit	Marks	Mins	Clock	Running total	Why it is this high up
1	Unitary / Hermitian check	V	05	08	0:08	5	Two minutes of work in the exam. Nothing on the paper pays better per minute.
2	Prove T is linear + find images	III	07	12	0:20	12	Q5b in all four papers. Zero calculus, zero matrices.
3	Least-squares solution	IV	10	20	0:40	22	10 marks, identical A and b twice , and it is only $A^T A x = A^T b$.
4	t^n -multiplication proof	I	07	15	0:55	29	4/4 papers, identical wording. Calculus on paper, pure recall in practice.
5	Heaviside piecewise + LT	II	07	20	1:15	36	Same function twice. Mechanical once you have the shift pattern.
6	$L\{\sin \sqrt{t}\} \rightarrow L\{\cos \sqrt{t}/\sqrt{t}\}$	I	04	12	1:27	40	Same statement twice. Four lines of differentiation.
7	The Unit I 2-mark definition	I	02	05	1:32	42	Five minutes for two marks, and it is in <i>both</i> options.
8	Orthogonal diagonalization	V	10	25	1:57	47	Same matrix 3/4. Longest item here, but you already know the eigenvalues.
9	$L^{-1}\{\log(\dots)\}$	II	04	12	2:09	52	Same expression twice. This is the row that crosses 50.
10	Span / basis / linear independence	III	06	12	2:21	53	Backs up Q5 so Unit III is safe whichever question appears.
11	Composition of transformations	III	06	12	2:33	58	Three 2×2 matrices multiplied. Opens up Q6.
12	Kernel & range + rank-nullity	III	07	15	2:48	59	Completes Q6 — now both Unit III questions are fully answerable.
— 3 HOURS UP. Everything below is overtime, in the order that pays best. —							
13	Transition matrix / change of basis	III	07	15	—	65	Biggest remaining jump: +6. If you can steal 15 minutes anywhere, steal them for this.
14	Positive-definite check	V	05	08	—	65	Three determinants. Pairs with step 1 to finish Unit V's 5+5.
15	Four fundamental subspaces	IV	07	15	—	65	Same matrix 3/4, one row reduction. Always in Q7.
16	Complete solution of $Ax = b$	IV	06	15	—	65	Same row-reduction skill you just used.
17	Unit II definitions + short inverse LT	II	06	15	—	65	Unit step, Dirac delta, and one partial-fraction inverse.

CUT. DO NOT OPEN THESE. Every one is real and several are 4/4 — but each is long, calculus-heavy, and the numbers change every paper, so three hours cannot buy them:
ODEs and simultaneous ODEs by Laplace • convolution theorem • improper integrals $\int_0^\infty$ • $L\{t e^{at} \text{ trig}\}$ and division by t • periodic functions / triangular wave • $(\sqrt{t+1}/\sqrt{t})^3$ • **QR factorization** • **SVD** • diagonalize and find A^n • Gram-Schmidt • orthogonal projection • Markov • quadratic forms • rotation-operator proof • PCA.
The four calculus items that survived (steps 4, 5, 6, 9) survived on your own rule: each has been set with **identical wording or identical values** in at least two papers.

IN THE HALL. You will not be able to answer every unit fully — that is fine and it is priced in. Rule: **write every part you know, in every unit, and never leave a unit blank.** A 5-mark property check you can do beats a 10-mark decomposition you half-remember. Answer the parts, not the question.

1 Unitary / Hermitian check

05 marks · Unit V · Q10a or Q10b (the 5+5 pair)

Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary? and the other version: is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian?

Method: write $A^* =$ conjugate of the transpose. **Hermitian** $\Leftrightarrow A^* = A$. **Unitary** $\Leftrightarrow A^*A = I$. **Both answers here are YES** — the 2×2 is unitary (and Hermitian too), and the 3×3 is Hermitian: diagonal is real and every $a_{ij} = \text{conj}(a_{ji})$.

2 Prove T is linear + find images

07 marks · Unit III · Q5b — all four papers, same slot

- (a) $T(x, y) = (3x+y, 2y, x-y)$; find images of $(1, 2)$ and $(2, -5)$.
 (b) $T: P_2 \rightarrow P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$; image of $5x^2+6x+1$.

Method (same both times): show $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and $T(k\mathbf{u}) = kT(\mathbf{u})$, then substitute. **Answers:** $T(1,2) = (5, 4, -1)$, $T(2,-5) = (1, -10, 7)$; $T(5x^2+6x+1) = 11x^2 + 1$.

3 Least-squares solution of $Ax = b$

10 marks · Unit IV · Q8b (or Q7b) — all four papers

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix} \quad \text{— identical in Jun 23 and Mar 24}$$

Method: solve the normal equations $A^T Ax = A^T b$. That is all.

Work it once now and memorise the shape: $A^T A = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$, $A^T b = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$, $\Rightarrow \hat{x} = (10, -6, 2)$.

4 Prove the t^n -multiplication rule

07 marks · Unit I · Q1c or Q2b — 4/4 papers, word for word

If $L\{f(t)\} = F(s)$, prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, n a positive integer.

Script — learn these four lines. $F(s) = \int_0^\infty e^{-st} f(t) dt$. Differentiate under the integral sign: $F(s) = \int_0^\infty (-t) e^{-st} f(t) dt = -L\{t f(t)\}$. Repeat n times $\rightarrow$ each pass drops one factor of $(-t)$. Finish by **induction on n** : assume true for n , differentiate once more, get $n+1$. State the base case $n=1$.

5 Heaviside piecewise + Laplace transform

07 marks · Unit II · Q3c — same function in Apr 23 and Mar 24

Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ in terms of the Heaviside function, and hence find its Laplace transform.

Step 1 — stack the jumps: $f(t) = t^2 + (4t - t^2)u(t-2) + (8 - 4t)u(t-4)$. **Step 2 — rewrite each bracket in $(t-a)$:** at $t = \tau+2$, $4t - t^2 = 4 - \tau^2$; at $t = \tau+4$, $8 - 4t = -8 - 4\tau$. **Step 3 — second shifting** ($L\{g(t-a)u(t-a)\} = e^{-as}G(s)$):
 $L\{f\} = 2/s^3 + e^{-2s}(4/s - 2/s^3) + e^{-4s}(-8/s - 4/s^2)$

6 Given $L\{\sin \sqrt{t}\}$, show the $\cos \sqrt{t}$ result

04 marks · Unit I · Q1b — same statement in Jun 23 and Mar 24

Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show that $L\{\frac{\cos \sqrt{t}}{\sqrt{t}}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$.

Method: notice $\frac{d}{dt}(\sin \sqrt{t}) = \frac{\cos \sqrt{t}}{2\sqrt{t}}$. So the target is $2 \times L\{\text{derivative}\}$. Apply $L\{f(t)\} = s F(s) - f(0)$, with $f(0) = \sin 0 = 0$.

That gives $2 \cdot s \cdot \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$. **Three lines.**

7 The Unit I 2-mark definition

02 marks · Unit I · Q1a and Q2a — you get one whichever you choose

“Write the Laplace transform of a periodic function” • “Define the Laplace transform of a function”

Periodic, period T : $L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$. **Definition:** $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$, provided the integral converges.
Write the formula, not a paragraph.

8 Orthogonally diagonalize

10 marks · Unit V · Q10c — same matrix in 3 of the 4 papers

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

You already know the answer — rehearse producing it. Eigenvalues **2, 3, 6** (all distinct, so the eigenvectors are automatically orthogonal — **no Gram-Schmidt needed**; say that in your answer).

Eigenvectors: $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ for 2, $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for 3, $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ for 6. Normalise by $\sqrt{2}, \sqrt{3}, \sqrt{6} \rightarrow$ those are the columns of P . Then $P^T A P = \text{diag}(2, 3, 6)$. Check all three dot products are 0 in your answer — it earns marks.

9 Inverse Laplace transform of a logarithm

04 marks · Unit II · Q3b — same expression in Apr 23 and Mar 24

Evaluate $L^{-1}\left\{\log \frac{s^2 + 1}{s(s+1)}\right\}$

Method: split the log $\rightarrow F(s) = \ln(s^2+1) - \ln s - \ln(s+1)$. Differentiate: $F'(s) = \frac{2s}{s^2+1} - \frac{1}{s} - \frac{1}{s+1}$. Invert term by term: 2 cos $t - 1 - e^{-t}$. Then use $L^{-1}\{F(s)\} = -t f(t)$:

answer = $\frac{1 + e^{-t} - 2 \cos t}{t}$ — this is the row that takes you past 50.

10 Span / basis / linear independence

06 marks · Unit III · Q5a — all four papers

Is $\left\{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}\right\}$ a basis of M_{22} ? or: do $\{(1,2,3), (-1,-1,0), (2,5,4)\}$ span R^3 ?

One method for every version: flatten each object into a coordinate vector (a 2×2 matrix becomes a 4-vector; a polynomial becomes its coefficient vector), stack them as the rows of a square matrix, and take the **determinant**. Non-zero $\Rightarrow$ independent $\Rightarrow$ basis / spans. Zero $\Rightarrow$ dependent. For “is $\mathbf{v}$ a linear combination of ...”, solve the system instead and show it is consistent.

11 Composition of matrix transformations

06 marks · Unit III · Q6a — all four papers

Determine the matrix for: reflection in the x -axis, then rotation through $\pi/2$, then contraction of factor $1/3$. Find the image of $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$. (the axis, angle, factor and point change every paper — the method does not)

Multiply RIGHT to LEFT: $M = k \cdot R_\theta \cdot \text{Ref}$. Reflection in x : $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. in y : $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. rotation: $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.
dilation/contraction by k : multiply the whole matrix by k . For $\pi/2$ the rotation is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then apply M to the point.

12 Kernel and range + verify rank-nullity

07 marks · Unit III · Q6c — all four papers

Determine the kernel and range of the transformation defined by $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$, and hence verify the rank-nullity theorem.

Method: row-reduce A . The **pivot columns of the original A** are a basis for the range; solving $Ax = \mathbf{0}$ gives a basis for the kernel. Here **rank = 2, nullity = 1**, and rank + nullity = 3 = number of columns. **Also be ready for the formula version** (e.g. $T(x,y) = (3x, x-y, y)$) — write its matrix first, then identical work.

OVERTIME — only if the clock allows

each row adds to the floor

13 Transition matrix / change of basis

07 marks · Unit III · Q6b — same bases in Sep 23 and Mar 24

$B = \{(1, 2), (3, -1)\}$, $B' = \{(3, 1), (5, 2)\}$ of R^2 . Find the transition matrix from B to B' ; if $[u]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $[u]_{B'}$.

Method: put the B vectors as the columns of Q and the B' vectors as the columns of P . The transition matrix is $Q^{-1}P$; then $[u]_{B'} = (Q^{-1}P)[u]_B$. One 2×2 inverse — **+6 to your worst-case score, the biggest single jump left**.

14 Positive-definite check

05 marks · Unit V · Q10a — same matrix in Apr 23 and Mar 24

Is $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite?

Method: leading principal minors. Here they are **1, 0, 0** — so it is **NOT positive definite**; it is positive *semi*-definite (row 3 = row 1, eigenvalues 0, 0, 6). **Say that explicitly** — students who just write “no” lose marks. Pairs with step 1 to close out Unit V's 5+5.

15 Four fundamental subspaces07 marks · Unit IV · **Q7 — always. Same matrix 3 of 4 papers.**

Find the dimension and basis for the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

Row 3 = row 1, so rank = 2 immediately. $\dim \text{col}(A) = \dim \text{row}(A) = \mathbf{2}$; $\dim \text{nul}(A) = 4 - 2 = \mathbf{2}$; $\dim \text{nul}(A^T) = 3 - 2 = \mathbf{1}$. Row-reduce once and read off a basis for each. One of the cheapest 7 marks on the paper.

16 Complete solution of $Ax = b$ + Unit II definitions

06+02 marks · Unit IV, II · Q7a · Q3a and Q4a

(a) Complete solution of $x+3y+3z = 1$, $2x+6y+9z = 5$, $-x-3y+3z = 5$.

(b) Define the unit step function (+ graph) • Define the Dirac-delta function (+ sketch).

(a) Row-reduce the augmented matrix, take any particular solution $\mathbf{x}_p$, add the null-space basis: write the answer as $\mathbf{x} = \mathbf{x}_p + \mathbf{c}_1 \mathbf{n}_1 + \dots$ (b) $u(t-a) = 0$ for $t < a$, 1 for $t \geq a$ — draw the step. $\delta(t-a)$ is the limit of a unit-area pulse, zero everywhere except $t = a$, with $\int \delta = 1$ — draw the spike arrow.

How this order was built: each of the 18 candidate topics was costed in minutes and scored by how many marks it adds to the *worst* of the four past papers, respecting the fact that you can only collect from one question per unit. The list is then greedy-ordered by worst-case marks per minute, so stopping at any row leaves you with the best score reachable in the time spent. Running totals are computed, not estimated. Full working: `coverage_audit.py` in the repo.